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Studierichting: Informatica
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$$\sin(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

Restterm (Lagrange):

$$R_{2n+1}(x) = \frac{x^{2n+2} \sin^{(2n+2)} c}{(2n+2)!}, -|x| < c < |x|$$

Voor $x = -0.7$ geldt

$$|R_{2n+1}(-0.7)| \leq \frac{0.7^{2n+2}}{(2n+2)!} \quad (\text{omdat } |\sin^{(k)} c| \leq 1)$$

Kies $n = 4$ zodat

$$|R_9(-0.7)| \leq \frac{0.7^{10}}{10!} < 10^{-7}$$

Daarmee

$$\begin{aligned} \sin(-0.7) &\approx \sum_{k=0}^4 (-1)^k \frac{(-0.7)^{2k+1}}{(2k+1)!} \\ &= -0.7 - \frac{(-0.7)^3}{3!} + \frac{(-0.7)^5}{5!} - \frac{(-0.7)^7}{7!} + \frac{(-0.7)^9}{9!} \\ &\approx -0.644217688 \end{aligned}$$

(eerste 7 decimalen gegarandeerd juist)

References